Acronal® EDGE 4750

Premium performance for Paint and Primer in One

BASF
We create chemistry

Acronal® EDGE 4750

Engineered to combine the best of performance requirements for primer and top coat.

Acronal EDGE 4750 is a high performance all acrylic latex designed for interior and exterior flat to semi-gloss paints. Acronal EDGE 4750 provides the foundation for premium performance with versatile formulation latitude. This latex unites the requirements of a primer and top coat, trapping existing stains through a combination of fast film formation and high opacity. Acronal EDGE 4750 further enhances a coating's durability by providing superior stain resistance.

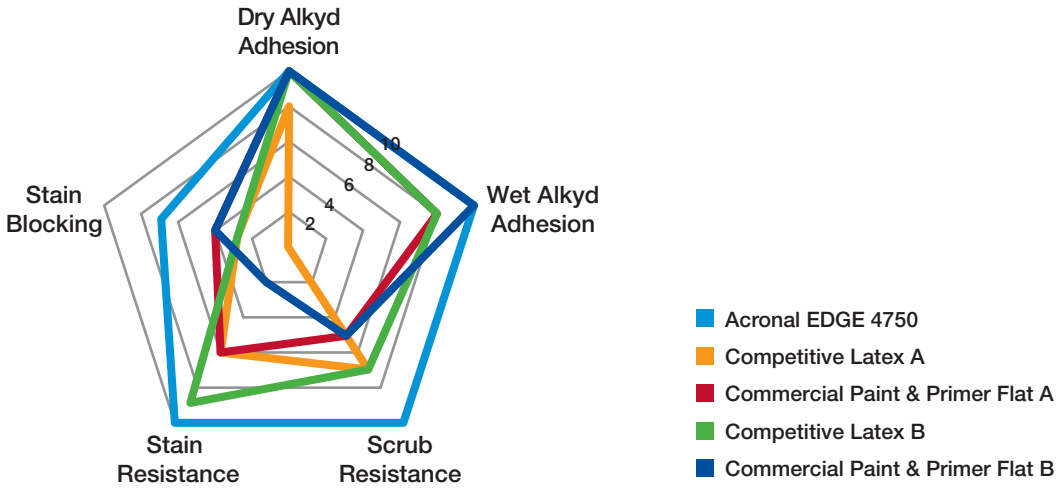
Features

- Outstanding stain resistance
- Built-in stain blocking
- Adhesion to multiple substrates
- Very good block resistance
- Excellent scrub resistance
- Zero VOC capable
- Excellent tannin blocking
- Very good dirt pick up resistance.

Properties

Solids content %	49.0 – 51.0
Solids content, Volume %	46.0 – 48.0
VOC content, weight %	< 0.2
VOC content, volume %	< 0.2
Viscosity	< 500 cps
pH cps	7.5 – 8.5
Specific gravity g/cm^3	ca. 1.06
Density lbs/gal	ca. 8.8
MFFT	ca. 15
Particle size nm	ca. 120

Performance of Acronal EDGE 4750 in an interior/exterior flat paint & primer formulation



	Acronal EDGE 4750	Competitive Latex A	Commercial Paint & Primer Flat A	Competitive Latex B	Commercial Paint & Primer Flat B
Dry Alkyd Adhesion	10	8	10	10	10
Wet Alkyd Adhesion	10	0	8	8	10
Scrub Resistance	10	7	7	5	5
Stain Resistance	10	6	9	6	3
Stain Blocking	7	3	3	4	4

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Stain Blocking

Acronal EDGE 4750 meets the performance demands for the paint and primer in one market.

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Competitive Latex

Stain Removal

These panels demonstrate the enhanced stain removal properties of Acronal EDGE 4750 in a flat formula.

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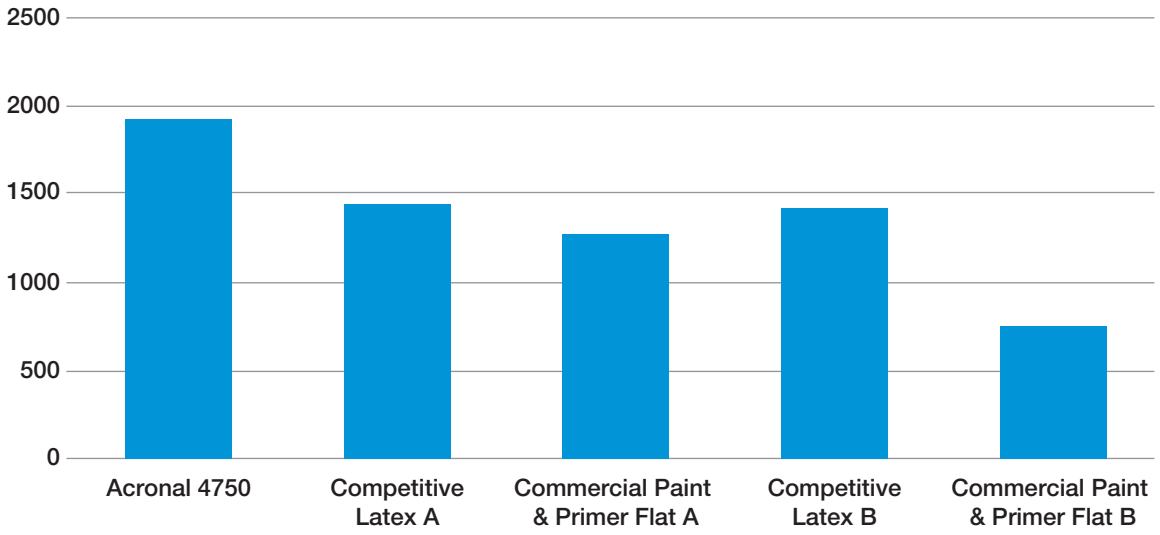
Competitive Latex

Scrub Resistance in Flat Paints



Scrub Resistance

Acronal EDGE 4750 has excellent durability and scrub resistance compared to competitive latexes and premium commercial paint and primer in one products.

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Formulation Guidelines

Coalescents

Texanol, added at 6.6% on binder solids, can give good low temperature film formation and performance. Non-volatile coalescents may require a higher loading. The use of reactive pigments such as zinc oxide may require special formulation considerations.

Efka PL 5651 or Optifilm 400 can be substituted for Texanol for reduced formula VOC.

Pigments

Both dry and slurry universal grades of titanium dioxide have been tested and work well with Acronal EDGE 4750. Choice of dry or slurry titanium dioxide can depend upon desired appearance, performance, and processing constraints. Titanium Dioxide slurry should be added in the letdown for optimal gloss development.

Through minimizing the use of high oil adsorption extenders along with maintaining low PVC, optimal stain removal can be achieved. For exterior performance, zinc oxide may be used for enhanced tannin blocking.

Dispersants/Surfactants

Ammonia carboxylic acid salt, such as Tamol 165A, and ammonium polyacrylate copolymer, such as Dispex CX4240, work well for pigment wetting and dispersing. Tamol 681 works well in dispersing zinc oxide. Fatty acid based dispersants are detrimental to stain resistance. Non-ionic surfactant, Hydropalat WE 3320, provides good wetting of the pigment and filler during dispersion preparation. It is suggested that less than 3 lbs/ 100 gal be used to provide good balance of wetting and hydrophilic stain resistance. Hydropalat WE 3320 can also be used to adjust rheology modifier efficiency.

Thickeners

Cellulosic thickeners should be limited in use for better flow and levelling. If needed for grind viscosity, a lower molecular weight HEC, such as Natrosol 250 HBR, may be used with very low impact on flow.

Attapulgite clay should be limited in use (<4 lbs/100gal) for optimal shelf stability. A small amount of delaminated kaolin, such as ASP NC-X1, can also reduce syneresis development and pigment settling.

For low-shear thickening, Newtonian HEUR thickeners, such as Rheovis PU 1251, should be used for the best flow and sag balance. In conjunction, Rheovis PU 1341 and Aquaflow NHS-310 work well for high-shear thickening.

Biocides

Iodopropyl buty carbamate (IPBC) is acceptable for mildew protection in formulas that may require mildew resistance. For exterior applications, Acticide MKW2 can protect the paint film from mildew, algae, and fungi attack. Formaldehyde or formaldehyde-donor preservative should not be used as they can adversely impact performance.

In-can preservatives, such as MIT, BIT, or CIT, and combinations of these, are adequate for long-term preservation.

Defoamers

While it is advised to avoid mineral oil defoamers, we have found that FoamStar ST-2412 works well to reduce foam in processing. FoamStar ST-2420 is an excellent defoamer for reducing brush and roller foam. FoamStar ST-2420 also reduces foam without decreasing gloss or introducing haze and film defects. Some ether-modified silicone oils have shown some compatibility issues and film defects upon incorporation.

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Suggested Formulations

Stain Resistant Flat 4750-003

raw materials	lbs	gallons
Grind		
Water	125.0	15.01
Natrosol 250 H4BR[1]	0.5	0.05
AMP 95[2]	1.0	0.13
Tamol 165A[2]	8.0	0.91
FoamStar ST 2412[3]	2.0	0.29
Acticide BW 20[4]	4.6	0.50
TiPure R706[5]	175.0	5.24
Minex 3[6]	125.0	5.75
ASP NC X-1[3]	25.0	1.16
Celite 281[7]	15.0	0.78
Attagel 50[3]	2.0	0.10
Grind Total	**483.1**	**29.92**
Grind for 15-20 minutes, then add Letdown		
Letdown		
Water	146.6	18.08
Foamstar ST 2420[3]	2.0	0.28
Acronal EDGE 4750[3]	425.0	48.08
Texanol[8]	14.0	1.77
Rheovis PU 1341[3]	16.0	1.83
Rheovis PU1251[3]	1.3	0.23
Acticide MKW2[4]	4.0	0.41
Letdown Total	**608.9**	**70.08**
Total	**1092.0**	**100.00**
PVC	36.52%	
VOC	44g/L	
Volume Solids:	36.58%	
Weight Solids:	51.93%	
KU:	100 - 110	
ICI:	1.0 - 1.5	
Gloss:		
60 degree:	5 Max	
85 degree:	5 Max	

Stain Resistant Semi-Gloss 4750-005

raw materials	lbs	gallons
Grind		
Water	70.0	8.40
Natrosol Plus 330[1]	0.3	0.03
Ammonium Hydroxide (28%)	0.5	0.07
Hydropalat WE 3320[3]	2.0	0.23
Dispex CX 4240[3]	2.0	0.21
FoamStar ST 2420[3]	2.2	0.31
Attagel 50[3]	4.0	0.20
Timed Hold 10 Minutes		
Minex 12[6]	22.0	10.10
Grind for 15-20 minutes, then add Letdown:		
Letdown		
Water	102.3	12.28
Acronal EDGE 4750[3]	500.0	56.52
Kronos 4311[10]	285.0	14.56
Timed Hold 10 Minutes		
Texanol[8]	16.0	2.02
FoamStar ED 2522[3]	2.0	0.24
Acticide BW20[4]	4.6	0.50
Acticide MKW2[4]	4.0	0.41
Aquaflow NHS-310[1]	25.0	2.88
Rheovis PU1251[3]	1.0	0.12
Total	**1042.9**	**100.00**
PVC	22.29%	
VOC	50g/L	
Volume Solids:	35.15%	
Weight Solids:	48.62%	
KU:	100 - 110	
ICI:	1.0 - 1.5	
Gloss:		
60 degree:	40 - 50	
85 degree:	80 min	

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Safety

General

The usual safety precautions when handling chemicals must be observed. These include the measures described in Federal, State and Local health and safety regulations, thorough ventilation of the workplace, good skin care and wearing of protective goggles.

Material Safety Data Sheet

All safety information is provided in the Material Safety Data Sheet for Acronal EDGE 4750.

Storage

Acronal EDGE 4750 should be stored in accordance with the “Handling and Storage of polymer dispersions” brochure. Technical information regarding the storage of BASF polymer dispersion products is available upon request. Product should not be allowed to freeze.

About the Dispersions & Resins Business

The Dispersions & Resins business of BASF develops, produces and markets a range of high-quality resins, additives, colorants and polymer dispersions worldwide. These raw materials are used in formulations for coatings and paints, printing and packaging products, construction chemicals, adhesives, fiberbondings, nonwovens, and paper manufacturing. With a comprehensive product portfolio and extensive knowledge of the industries we serve, our customers benefit from innovative and sustainable solutions to help them advance their formulations through chemistry. For further information about the Dispersions & Resins business in North America, please visit http://www.basf.us/dpsolutions

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